

GATE Practice Questions:

Functions & Recursion - Complete Solutions

Question 1 (GATE 2016)

Question: Consider the following C function:

```
int fun(int n) {  
    int i, j, sum = 0;  
    for (i = 1; i <= n; i++)  
        for (j = 1; j <= i; j++)  
            sum++;  
    return sum;  
}
```

The return value of fun(5) is:

(A) 10 (B) 15 (C) 21 (D) 25

Answer: (B) 15

Question 2 (GATE 2016)

Question: Consider the following C program:

```
#include <stdio.h>  
void swap(int *x, int *y) {  
    static int *temp;  
    temp = x;  
    x = y;  
    y = temp;  
}  
int main() {  
    int r = 10, s = 20;  
    swap(&r, &s);  
    printf("%d %d", r, s);  
    return 0;  
}
```

The output of the program is:

(A) 20 10 (B) 10 20 (C) 20 20 (D) 10 10

Answer: (B) 10 20

Question 3 (GATE 2015)

Question: Consider the following C function:

```
int fun(int n) {  
    static int i = 1;  
    if (n >= 5) return n;  
    n = n + i;  
    i++;  
    return fun(n);  
}
```

The value returned by fun(1) is:

(A) 5 (B) 6 (C) 7 (D) 8

Answer: (C) 7

Question 4 (GATE 2014)

Question: Consider the following C function:

```
int f(int n) {  
    static int r = 0;  
    if (n <= 0) return 1;  
    if (n > 3) {  
        r = n;  
        return f(n-2) + 2;  
    }  
    return f(n-1) + r;  
}
```

What is the value of f(5)?

(A) 5 (B) 7 (C) 9 (D) 18

Answer: (D) 18

Question 5 (GATE 2011)

Question: Consider the following recursive C function:

```
unsigned int foo(unsigned int n, unsigned int r) {  
    if (n > 0) return ((n % r) + foo(n / r, r));  
    else return 0;  
}
```

What is the return value of foo(513, 2)?

(A) 2 (B) 3 (C) 4 (D) 5

Answer: (A) 2

Question 6 (GATE 2008)

Question: What does the following C function perform?

```
int fun(int n) {  
    int x = 1, k;  
    if (n == 1) return x;  
    for (k = 1; k < n; ++k)  
        x = x + fun(k) * fun(n - k);  
    return x;  
}
```

(A) Computes the n-th Fibonacci number (B) Computes the n-th Catalan number
(C) Computes the n-th Factorial number (D) None of the above

Answer: (B) Computes the n-th Catalan number

Question 7 (GATE 2007)

Question: Choose the correct output for the following C program:

```
#include <stdio.h>  
int f(int *a, int n) {  
    if (n <= 0) return 0;  
    else if (*a % 2 == 0) return *a + f(a + 1, n - 1);  
    else return *a - f(a + 1, n - 1);  
}  
int main() {  
    int a[] = {12, 7, 13, 4, 11, 6};  
    printf("%d", f(a, 6));  
    return 0;  
}
```

(A) -9 (B) 5 (C) 15 (D) 19

Answer: (C) 15

Question 8 (GATE 2005)

Question: Consider the following C function:

```
int Count(int n) {  
    static int d = 1;  
    printf("%d ", n);  
    printf("%d ", d);  
    d++;  
    if (n > 1) Count(n - 1);  
    printf("%d ", d);  
}
```

What is the output of Count(3)?

- (A) 3 1 2 2 1 3 4 4 4 (B) 3 1 2 2 1 3 4
(C) 3 1 2 2 1 3 2 2 2 (D) 3 1 2 2 1 3 3 3 3

Answer: (A) 3 1 2 2 1 3 4 4 4

Question 9 (GATE 2004)

Question: Consider the following C function:

```
void exchange(int *a, int *b) {  
    int temp;  
    temp = *a;  
    *a = *b;  
    *b = temp;  
}
```

If the call is exchange(&x, &y), then this parameter passing mechanism is known as:

- (A) Call by value (B) Call by reference (C) Call by value-result (D) None of the above

Answer: (A) Call by value

Question 10 (GATE 2012)

Question: What is the output printed by the following C program?

```
#include <stdio.h>  
void printxy(int x, int y) {  
    int *ptr;
```

```
x = 0;
ptr = &x;
y = *ptr;
*ptr = 1;
printf("%d, %d\n", x, y);
}
int main() {
    printxy(25, 25);
    return 0;
}
```

(A) 0, 0 (B) 1, 0 (C) 0, 1 (D) 1, 1

Answer: (B) 1, 0

Question 11 (GATE 2017)

Question: Consider the following C code:

```
#include <stdio.h>
int jumble(int x, int y) {
    x = 2 * x + y;
    return x;
}
int main() {
    int x = 2, y = 5;
    y = jumble(y, x);
    x = jumble(y, x);
    printf("%d\n", x);
    return 0;
}
```

The value printed by the program is:

(A) 12 (B) 26 (C) 54 (D) 62

Answer: (B) 26

Question 12 (GATE 1999)

Question: Consider the following recursive function:

```
int recursive(int n) {  
    if (n == 1) return 1;  
    else return (recursive(n - 1) + n);  
}
```

What does this function calculate for a positive integer n?

(A) n^2 (B) $n!$ (C) $n(n+1)/2$ (D) 2^n

Answer: (C) $n(n+1)/2$

Question 13 (GATE 2018)

Question: Consider the following C program:

```
#include <stdio.h>  
void f(int *p, int *m) {  
    p = m;  
    *p = 2;  
}  
int main() {  
    int i = 0, j = 1;  
    f(&i, &j);  
    printf("%d %d\n", i, j);  
    return 0;  
}
```

The output of the program is:

(A) 0 1 (B) 0 2 (C) 2 1 (D) 2 2

Answer: (B) 0 2

Question 14 (GATE 2019)

Question: Consider the following C function:

```
int fun(int n) {  
    if (n <= 1) return 1;  
    if (n % 2 == 0) return fun(n / 2);  
    return fun(n / 2) + fun(n / 2 + 1);  
}
```

The value of fun(11) is _____.

Answer: 5

Question 15 (GATE 2020)

Question: Consider the following C function:

```
int f(int n) {  
    if (n <= 1) return 1;  
    return f(n-1) + f(n-2);  
}
```

How many activation records (stack frames) will be created in total for the function call $f(4)$?

(A) 5 (B) 7 (C) 9 (D) 8

Answer: (C) 9